

NGUYEN GIA HUY

Data Engineer

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PROFESSIONAL SUMMARY

Data engineer with nearly three years of software engineering experience, most deeply in production data pipelines on AWS. At Vietlink (KDDI group), designed, built, and operated the ETL platform behind **au Denki**, an electric utility service with millions of users: serverless and containerized processing on Lambda and ECS, S3 and Athena for analytical querying, RDS for relational data, Terraform and AWS SAM for infrastructure, and CloudWatch and DataDog for monitoring. Raised batch throughput by **150%** and migrated search analytics from Elasticsearch to OpenSearch. At Vikki Digital Bank, built the Golang account migration system that cut a migration run from **2.5 hours to 27 minutes**.

TECHNICAL SKILLS

Data Engineering:	ETL pipeline design, data ingestion, data transformation, batch processing, parallel processing
Migration & Search:	Account data migration, search indexing, Elasticsearch to OpenSearch migration
AWS Data Services:	Amazon S3, Amazon Athena, AWS Lambda, Amazon ECS, Amazon RDS
Languages & Query:	SQL, Python, Golang
Databases & Search:	PostgreSQL, OpenSearch, Elasticsearch, ClickHouse, MongoDB, Redis
Streaming & Messaging:	Kafka, RabbitMQ
Infrastructure:	Terraform, AWS SAM, Docker, Kubernetes
Monitoring:	CloudWatch, DataDog, Prometheus, Grafana, OpenTelemetry, ELK

WORK EXPERIENCE

Middle Software Engineer – Core Banking Data & Migration 03/2026 – Present

Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Built the **account migration system** for the core banking platform in **Golang**, reducing a migration run from **2.5 hours to 27 minutes (5x faster)**.
- Ran migration processing on AWS, combining containerized workloads on **ECS** with serverless functions on **Lambda** and persisting migration data in **RDS**.
- Implemented interest accrual plus penalty and late-fee calculations for the **Lending** and **Overdraft** products in Python Smart Contracts on Thought Machine Vault Core.
- Implemented **repayment and settlement** logic, where every calculated amount flows into customer balances on the core ledger.
- Integrated Vault Core with external backend services, supporting core banking operations and transaction workflows across systems.
- Rebuilt the simulation testing framework entirely in Golang, replacing the Gauge-based framework and cutting execution from **24 minutes to just over 50 seconds (20x faster)**.
- Deployed a new Vault Core version on AWS (EKS, RDS, ECR, ELB) and built its CI/CD pipeline for automated build, test, and deployment.

Data Engineer (Software Engineer) 04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

Data Pipeline Architecture

- Designed, developed, and operated the **ETL pipelines** of au Denki's data platform, transforming and processing large-scale utility data for downstream analytics and operational workloads.
- Architected the end-to-end data flow across AWS: processing on **Lambda** and **ECS**, datasets stored in **S3**, SQL analytics through **Athena** on top of S3, and **RDS** as the relational store.
- Combined serverless processing on AWS Lambda with containerized batch jobs on Amazon ECS, so each pipeline stage runs on the execution model that suits its workload.
- Built the ingestion and transformation stages that turn raw utility data into structured, analytics-ready datasets for reporting and operational use.

Batch Processing Optimization

- Optimized batch-oriented data processing workflows, increasing overall pipeline throughput by **150%**.
- Redesigned the batch pipelines around **parallel execution**, processing independent units of work concurrently to make better use of available compute.
- Reworked **data access patterns** inside pipeline stages so that reads and writes against S3 and RDS are more efficient per batch.

Search and Analytics Platform

- Migrated the search and analytics platform from **Elasticsearch to OpenSearch**, modernizing the search infrastructure behind analytical queries.
- Redesigned the **data ingestion flows** that feed search, improving stability and keeping indexed data consistent with its source.
- Restructured **indexing flows** during the migration to improve query performance for search and analytics workloads.

Data Reliability and Operations

- Provisioned and maintained the data platform's infrastructure as code with **Terraform**, keeping deployments of AWS data workloads consistent across releases.
- Defined and deployed the serverless pipeline components with **AWS SAM**.
- Implemented pipeline monitoring in **CloudWatch** and **DataDog** for pipeline health, failure detection, and operational anomaly detection on a platform serving millions of users.

Software Engineer – Cloud Platform Infrastructure

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Designed and operated VNPAY Cloud's private cloud on **OpenStack**, running **50+ production Kubernetes clusters** for fintech and enterprise customers.
- Automated Kubernetes cluster provisioning and scaling with **Cluster API** and carried out zero-downtime upgrades of production clusters.
- Automated about **80%** of data center provisioning and infrastructure configuration processes, significantly reducing manual operational effort.
- Integrated and operated **Kong Gateway** for a microservices ecosystem of **100+ services**, enforcing service-level and network-level access control.
- Implemented multi-tenant authentication and authorization with **Keycloak** and RBAC, ensuring tenant isolation across **20+ customer organizations**.

Blockchain Developer – On-chain Data Ingestion

06/2023 – 09/2023

EsoiLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built **event-driven ingestion services** that capture and process on-chain events and synchronize them into off-chain databases and APIs.
- Integrated the services with **5+ blockchain networks**, covering EVM chains (Ethereum, BSC, Polygon) and Move-based chains (Aptos, Sui).
- Kept transaction handling consistent and preserved **data integrity** as events moved from multiple chains into off-chain storage.
- Developed and reviewed **15+ smart contracts** for DeFi and NFT use cases in Solidity and Move.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built the backend around **Kafka** for messaging, with **MongoDB** and **Redis** as data stores, exposed through gRPC-Gateway.
- Sustained up to **1.5K requests/second** at **80–100 ms** API latency on that stack.
- Implemented an observability stack with OpenTelemetry, Prometheus, Grafana, ELK, and **ClickHouse**, plus multi-cluster failover, achieving **99.95% uptime**.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on **LiveKit**, integrating OpenAI models for live transcription, voice translation, and meeting summarization.
- Designed the **real-time translation pipeline** and optimized conversational context handling to improve translation accuracy and coherence.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera